نقل الكهرباء
National Grid SA

ORIGINAL DOCUMENT APPROVED December 17, 2007

REVISIONS		
No.	Date	DESCRIPTION
1	January 21, 2013	MODIFICATION TO SECTION 7.0 AND GENERAL MODIFICATIONS
2	May 22, 2017	UPDATED DATA INCLUSION AND MODIFICATION OF DIFFERENT CLAUSES
3	November 15, 2018	Modification of different clauses, inclusion of new standards and changes in details for some equipment

TES-P-119.19

Revision No. 03

SUBSTATION BUILDING AND SITE DEVELOPMENT

PROPRIETARY INFORMATION

This document contains proprietary information developed by and for exclusive use of National Grid Saudi Arabia. Your acceptance of the document is an acknowledgement that it must be used for the identified purpose/application and during the period indicated. It cannot be used or copied for any other purposes nor released to others without prior written authorization of National Grid Saudi Arabia. National Grid Saudi Arabia will not be responsible for misuse and/or misapplication, and any harm resulting there from. National Grid Saudi Arabia also reserves the right to take any necessary actions to protect its interest against unauthorized use.

ORIGINATED BY	CERTIFIED BY	APPROVED BY
Engr. Majed A. Al-Suhaibani Division Manager, Prequalification & Support Facilities	**Engr. Suliaman N. Al-Dosari** Department Manager Standards & Specifications	**Engr. Waleed A. Al-Saadi** Vice President Engineering

Date of Approval: November 15, 2018

TRANSMISSION ENGINEERING STANDARD

TABLE OF CONTENTS

1.0 SCOPE

2.0 CROSS REFERENCES

3.0 APPLICABLE CODES AND STANDARDS

4.0 GENERAL

4.1 Purpose of Substation Building
4.2 Environmental Considerations
4.3 Equipment Limitations
4.4 Physical Layout and Clearances

5.0 EQUIPMENT ARRANGEMENT

5.1 Relay and Control Panels
5.2 Metalclad Switchgear
5.3 Station Battery
5.4 Battery Charger
5.5 Auxiliary Panels
5.6 Communication Equipment
5.7 SCADA Equipment
5.8 Air Conditioning Equipment
5.9 69kV Air Insulated Switchgear (AIS)
5.10 69/110/115/132/230/380kV Gas Insulated Switchgear (GIS)
5.11 Fire Suppression and Detection Equipment

6.0 HEATING, VENTILATION AND AIR CONDITIONING (HVAC) SYSTEMS REQUIREMENTS

6.1 General
6.2 HVAC Design Parameters
6.3 HVAC Control System

7.0 CIVIL/STRUCTURAL REQUIREMENTS

7.1 Building Design Criteria
7.2 Substation Building
7.3 Roads and Walkways
7.4 Site Development
7.5 Fencing and Boundary Wall
7.6 Substation Signboard
7.7 Protective Crash Barriers
7.8 Electric Overhead Travelling Crane in GIS Buildings
7.9 Jib Crane
7.10 Gate House
7.11 Cable Tunnel

8.0 LIGHTING AND RECEPTACLE REQUIREMENTS

9.0 LIGHTNING PROTECTION REQUIREMENTS

10.0 GROUNDING REQUIREMENTS

11.0 FIRE AND LOSS PREVENTION REQUIREMENTS

1.0 SCOPE

This Transmission Engineering Standard (TES) covers the criteria and parameters for the design of substation building structure and the spaces requirements of the equipment in substation building. It also includes substation site development requirements such as roads and walkways, surfacing, drainage system including surface drainage and boundary wall/fencing, sign board and gate house.

2.0 CROSS REFERENCES

This Engineering Standard Specification shall always be read in conjunction with Latest Revisions of National Grid Saudi Arabia General Specification TES-P-119.01, Titled "Introduction to Substation Design Standards" and 01-TMSS-01, titled "General Requirements For All Equipment/Materials", which shall be considered as an integral part of this TES.

3.0 APPLICABLE CODES AND STANDARDS

Unless specified otherwise in this section, design, materials, manufacturer, workmanship and testing of all foundation works shall comply with an approved standard. All substation provided under this specification shall conform to the applicable codes and standards of:

The latest edition or revision of these approved standards shall apply.

3.1 Company Standards

3.1.1 Transmission Material Standard Specification

01-TMSS-01 General Requirements for All Equipment/Materials

38-TMSS-03 Remote Terminal Unit

70-TMSS-02 Chainlink Fence

70-TMSS-03 Normal Weight Ready-Mixed Portland Cement Concrete

73-TMSS-01 Air Conditioning Unit, Package

73-TMSS-02 Package Air Cooled Water Chiller with Screw-Type Compressor

73-TMSS-03 Air Handling Units, Chilled Water Type

3.1.2 Transmission Engineering Specification

TES-P-103.04 Storage Battery Installation

TES-P-107.01	Substation Automation System
TES-P-107.02	Supervisory Control and Data Acquisition System for Transmission Substations with Substation Automation System (SAS).
TES-P-119.07	Shielding
TES-P-119.09	Busbars and Supporting Structures
TES-P-119.10	Grounding
TES-P-119.20	LV Cable Raceways & Cable Installation Practices
TES-P-119.21	Fire & Loss Prevention and Security Requirements
TES-P-119.25	Lighting and Receptacles for Substation Yard and Building
TES-P-119.27	Supervisory Control and Data Acquisition (SCADA)
TES-P-119.30	Auxiliary Power Systems (AC/DC)
TES-P-122.05	Transmission Structures
TES-H-107.01	Painting
TES-K-100.01	Heating, Ventilation and Air Conditioning System Design
TES-T-111.02	Communication Facility Grounding
TES-T-111.03	Communication Facility Lighting and Electrical Protection
TES-Q-113.01	Design Criteria for Precast and Prestressed Concrete Structures
TES-S-101	Safety and Security Standards

3.1.3 Transmission Construction Standards

TCS-K-100.01	HVAC Systems and Equipment Installation
TCS-Q-113.02	Earthworks
TCS-Q-113.03	Cast in Place Concrete
TCS-Q-113.04	Precast Concrete Works

TCS-Q-113.05 Reinforced Concrete Masonry

TCS-Q-113.06 Concrete Rehabilitation

3.1.4 Transmission Standard Drawings

TA-800009 Steel Doors, Frames, and Hardware for SSD/7 Security Requirements

TA-800012 Precast Concrete Boundary Wall (Plan, Elevation & Details)

TA-800013 Boundary Walls for Substations Located in Urban Areas

TA-800023 Door Schedule and Details

TA-800024 Finish Schedule/ Miscellaneous Details

TA-800039 Plans, Elevations & Sections for Gatehouse

TA-800040 Foundation, Roof Plan & Details for Gatehouse

TA-800041 Miscellaneous Details for Gatehouse

TA-800049 Access Cage Ladder

TA-800111 Substation Signboard at Gate House

TA-800133 Removable Crash Barrier, Location Plan and Detail in Substation

TA-800134 Septic Tank Installation Details For Non-Traffic Area

TA-800135 Disposal System Layout & Seepage Pit Installation Details

TA-800137 Sanitary Sewer Manhole Precast Concrete Plan, Section & Details

TA-800138 Drainage Sump, Drainage Channel And Sewer Manhole Cover Details

TA-800144 Precast Boundary Wall Type-II (Typical Layout Plan, Elevation, Sections And Details)

TB-800079 PVC Conduit Installation Concrete Encased (Vital Communication)

TC-800003 Signboard on Substation Building

3.2 SASO - Saudi Arabian Standards Organization

SSA 2	Steel Bars for the Reinforcement of Concrete
SSA 14	Pipes for Potable Water of Unplasticized Plastic (Poly Vinyl Chloride)
SSA 15	Methods of Testing Pipes for Potable Water of Unplasticized Plastic (Poly Vinyl Chloride)
SSA 142	Physical and Mechanical Testing Methods of Portland Cement
SSA 143	Portland Cement -Ordinary & Rapid Hardening
SSA 378	Aggregates from Natural Sources for Concrete
SSA 690	Test Method of Steel Bar for Reinforcement of Concrete

3.3 International Standards

NFPA 70	National Electrical Code (NEC)
NFPA 72	National Fire Alarm and Signaling Code
NFPA 80	Standards for the Fire Doors and Other Opening Protective's
NFPA 101	Life Safety Code
NFPA 220	Standard on Types of Building Construction
ACI 318/ ACI 318M	Building Code Requirements for Reinforced Concrete
ACI SP-17	The Reinforced Concrete Design Handbook
ACI SP-66	Detailing Manual
ACI 216.1M	Code Requirements for Determining Fire Resistance of Concrete and Masonry Construction Assemblies
ANSI C2	National Electrical Safety Code
ASCE/SEI-7	Minimum Design Loads for Buildings and other Structures
ASCE MOP	74-Guidelines for Electrical Transmission Line Structural Loading

ASCE MOP 113 - Substation Structure Design Guide

ANSI/ASME B16.3	Malleable Iron Threaded Fittings Classes 150 and 300
ANSI/ASME B16.18	Cast Copper Alloy Solder Joint Pressure Fittings
ANSI A 112.6.3	Floor Drains & Trench Drains
ANSI/ ASME A112.19.2M	Vitreous China Plumbing Fixtures and Hydraulic Requirements for Water Closets and Urinals
ANSI/ASME A112.18.1	Plumbing Supply Fittings
ANSI/ASME A112.19.3M	Stainless Steel Plumbing Fixtures
ISEA Z 358.1	Emergency Eyewash and Shower Equipment
ANSI/ASTM D2729	Specification for Poly Vinyl Chloride (PVC) Sewer Pipe and Fittings
ASTM A53	Specification for Pipe Steel, Black and Hot Dipped, Zinc-coated, Welded and Seamless
ASTM A167	Stainless and Heat-Resisting Chromium-Nickel Steel Plate, Sheet and Strip
ASTM A615	Specification for Deformed and Plain Carbon Steel Bars for Concrete Reinforcement
ASTM A653 / A653M	Steel Sheet, Zinc-Coated (Galvanized)or zinc-Iron Alloy-Coated (Galvannealed) by the Hot-Dip Process
ASTM B88	Specification for Seamless Copper Water Tube
ASTM B62	Specification for Composition Bronze or Ounce Metal Castings
ASTM B209	Aluminum Alloy Sheet and Plate
ASTM C423	Test Method for Sound Absorption and Sound Absorption Coefficients by Reverberation Room Method
ASTM C457	Test Method for Microscopical Determination of Parameters of the Air-Void System in Hardened Concrete

ASTM C635	Metal Suspension Systems for Acoustical Tile and Lay-In Panels Ceilings
ASTM C636	Installation of Metal Ceiling Suspension Systems for Acoustical Tile and Lay-In Panels
ASTM D698	Moisture-Density Relation of Soils
ASTM D1194	Test for Bearing Capacity of Soil for Static Load on Spread Footings
ASTM D1557	Test Method for Laboratory Compaction Characteristics of Soil Using Modified Effort ((56,000 ft.-lb/ft.) (27,00 kN-m/m)
ASTM D1785	Specification for (Poly Vinyl Chloride) Plastic Pipe, Schedules 40, 80 and 120
ASTM D1998	Polyethylene Upright Storage Tanks
ASTM D2049	Test for Relative Density of Cohesionless Soils
ASTM D2487	Practice for Classification of Soils for Engineering Purposes (Unified Soil Classification System).
ASTM D2564	Specification for Solvent Cements for (Poly Vinyl Chloride) Plastic Piping System
ASTM D2855	Practice for Making Solvent - Cemented Joints with Polyvinyl Chloride (PVC) Pipe and Fittings.
ASTM D4253	Test Method for Maximum Index Density and Unit Weight of Soils and Calculation of Relative Density.
ASTM D4254	Test Method for Minimum Index Density and Unit Weight of Soils and Calculations of Relative Density.
ASTM E84	Surface Burning Characteristics of Building Materials
ASTM E90	Laboratory Measurement and Airborne Sound Transmission Loss of Building Partitions
ASTM E119	Methods of Fire Tests of Building Construction and Materials
ASTM 615M	Specification for Deformed and Plain Billet- steel Bars for Concrete Reinforcement (Metric)
AMCA 211	Certified Ratings Program - Product Rating Manual for Fan Air Performance

AMCA 311	Certified Sound Ratings Program for Air Moving Devices
IEEE 691	Guide for Transmission Structure Foundation Design and Testing
SBC	Saudi Building Code
UMC	Uniform Mechanical Code
UPC	Uniform Plumbing Code
IBC	International Building Code
IPC	International Plumbing Code
HCIS	High Commission for Industrial Security

3.4 Alternative Codes and Practice

The supplier/contractor may propose alternative codes and standards, provided it is demonstrated that they give an equivalent degree of qualities equal or more than the referenced codes and standards. Acceptability of any alternative code or standard is at the discretion of National Grid Saudi Arabia.

3.5 Precedence of Codes and Standards

In cases of conflict between this specification and any of the referenced codes and standards, the following order of precedence shall apply:

- This standard
- SASO Codes and Standards
- Other referenced codes and standards including Safety requirements
- Acceptable Alternative codes and standards

4.0 GENERAL

4.1 Purpose of Substation Building

The purpose of a substation building is to protect the equipment installed indoors, with necessary clearances and working space for the safe operation and maintenance of such equipment.

4.2 Environmental Considerations

All equipment/material and devices and their specific ratings shall be designed for satisfactory operation in environment as stated in TES-P-119.02.

4.3 Equipment Limitations

Indoor high voltage equipment shall be limited to 69kV conventional equipment and 69/110/115/132/230/380kV and above Gas Insulated Switchgear (GIS).

4.4 Physical Layout and Clearances

The size of the substation building shall provide adequate working space and electrical clearances for all indoor equipment. Wherever possible, similar equipment shall be located adjacent to each other, as follows:

4.4.1 Metal Enclosed Switchgear cubicles (See details of clearances in 5.2 below)
4.4.2 Control panels, AVR/RTCC Panels, Relay Panels, Annunciator Panels, Fault Recorder Panels, SOE/RTU Panels, and Substation Automation etc.
4.4.3 Battery Charger Panels, DC Distribution Panels, etc.
4.4.4 LV AC Distribution panels, Lighting Panels, etc.
4.4.5 Fire Protective Signaling Panels, etc.
4.4.6 GIS (Gas Insulated Switchgear) and AIS (Air Insulated Switchgear)
4.4.7 Station Batteries, Communication Batteries, etc.
4.4.8 Communication equipment
4.4.9 HVAC equipment
4.4.10 Fire Suppression and Detection Equipment with all related accessories.

Indoor equipment space allocation and space allowance shall be made for known or likely future expansion.

5.0 EQUIPMENT ARRANGEMENT

5.1 Relay and Control Panels

Relay and Control Panels shall be housed in the Control Room. A minimum clearance of 1800 mm shall be provided between the row of panels and from wall/column face which is nearest to the panel. Panels associated with one feeder or transformer and similar panels shall preferably be grouped together based on their functionality.

5.2 Metal Enclosed MV Switchgear

All section of Metalclad switchgear shall be housed in one room. The dimensions of the metalclad switchgear cubicle and space required for the removal of a circuit breaker shall vary depending on the design and rating of the metalclad switchgear or as per the recommendation of the manufacturer.

a. The minimum clearance between 13.8 & 33 kV MV switchgear panels and adjacent side walls shall be 1 Meter.

b. The minimum clearance between front of MV switchgear panels and wall shall be 2.5 Meters (For 13.8 kV Switchgear) and 3.5 Meters (For 33 kV Switchgear).

c. The minimum clearance between back side of 13.8 & 33 kV MV switchgear panels and wall shall be 1.8 meters (Panel Without Rear Cable Box) and 1.8 Meters (Panel With Rear Cable Box).

d. Provision of space and arrangement shall be allowed for future extension by at least one cubicle at each ends sections (A&C) of the 13.8 & 33 kV MV Switchgear line-up. The related work shall be considered to meet this requirement.

5.3 Station Battery

5.3.1 A separate battery room conforming to TES-P-103.04 shall be provided. Additional space shall be allocated for communication battery and future expansion, whenever required.

5.3.2 The battery room shall be sized according to battery bank size, layout of racks and as to provide a clearance of at least 1.5 meters between any battery and the eye and skin wash equipment.

5.3.3 The battery room shall be provided with a trapped and vented floor drain to serve eye and skin wash facilities and wash down of electrolyte spills. A hose bib of 20mm diameter shall be provided for wash down of electrolyte spills. The floor drain shall be connected to a neutralization pit, before connecting to the sanitary system.

All piping in the upstream of the neutralization pit**,** the vents and the neutralization pit shall be acid resistant. In remote areas, where sewer or drainage line close by is not available, the floor drain in the battery room shall be primed trap, un-vented and piped to a dry sump. Acid resistant portable water facilities or gravity water storage tank with piping shall be provided for eye and skin washing and flushing electrolyte spills inside the battery room.

Battery room shall be provided with proper Air ventilation to ensure diffusion of gases from the battery to prevent the accumulation of an explosive mixture.

5.4 Battery Charger

5.4.1 Battery charger shall be housed in control room for grid substations up to 132kV system voltage unless otherwise specified in SOW/TS. For BSP's, 230kV and 380kV substations, the same shall be housed in separate room called "Auxiliary AC/DC Room". The charger shall be mounted upright. Battery charger for communication system shall be installed in the communication room.

5.4.2 Chargers are usually constructed so that all components are readily accessible from the front. The manufacturer's recommendations for minimum working space for maintenance shall be taken into consideration; and the working space shall permit full opening of all doors. Nevertheless, the working space shall not be less than 1.0 m.

5.4.3 When chargers are located near other equipment, proper care shall be taken that its ventilation openings are not obstructed.

5.5 Auxiliary Panels

5.5.1 Auxiliary AC/DC Panels shall be housed in control room for grid substations up to 132kV system voltage unless otherwise specified in SOW/TS. For BSP's, 230kV and 380kV substations, the same shall be house in separate room called "Auxiliary AC/DC Room". LV AC distribution panels, DC distribution panels for station DC system, AC lighting panels, auto-transfer switch panels, intrusion alarm, fire alarm and annunciator panels and any other auxiliary panels which do not require back access shall be surface mounted on the interior face of the building wall, unless free-standing type is required. A minimum spacing of 150mm shall be maintained between panels to facilitate conduit connections. DC distribution panel for communication system shall be installed in the communication room.

5.5.2 Working space around auxiliary panels shall be provided in accordance with NFPA 70, Article 110-16, and “Working Space about Electric Equipment" (600 Volts Nominal or Less). Nevertheless, for maintenance purposes the working space shall not be less than 1.0 m.

5.6 Communication Equipment

5.6.1 A separate communication room with size sufficient enough to accommodate all required equipment for full substation development shall be allotted for communication equipment, fiber optic terminal, carrier equipment, and relaying and microwave equipment.

5.6.2 Cabinets and racks for communication equipment shall be of the following types:

a. Swinging-Rack Cabinet

This type can be located with its back directly against the building wall or placed side by side and shall provide convenient access to the rear of the chassis mounted on the rack. Typical dimensions of this type of cabinet are: Height 2200mm, Length 850mm and Width 650mm.

b. Fixed Equipment Rack

This type of equipment racks shall be floor mounted and installed side by side. Back access may be needed depending on equipment design. Typical height is 2200mm.

5.6.3 Manufacturer's dimensional data for a particular installation shall be used in all cases, in allocating minimum room space.

5.7 SAS/SCADA Equipment

5.7.1 Substation Automation System (SAS) equipment (Control/Protection IEDs, Ethernet switches, display units and HMI etc.) shall be installed in control rooms as per TES-P-107.01. The SAS networking panels shall be so installed that they could have access and working space from both (front & rear) sides.

5.7.2 SCADA equipment (viz. RTU, IFC and Transducers) shall be installed in the control room as per 38-TMSS-03 and TES-P-119.27. Manufacturer's dimensional data for a particular installation shall be used in all cases.

5.8 Heating, Ventilation and Air Conditioning Equipment

All HVAC equipment shall be designed and installed as per TES-K-100.01 and TCS-K-100.01 respectively.

All HVAC control panels shall be installed in the control room.

5.9 69kV Air Insulated Switchgear (AIS)

69kV Air Insulated Switchgear (AIS) shall be housed in a separate room. Physical layout and space requirements shall be per Scope of Work/Technical Specification (SOW/TS) and other relevant chapters of TES-P-119. Equipment arrangement for other type of bus configurations not specified in TES-P-119 shall be subject for review on a case to case basis.

5.10 69/110/115/132/230/380kV Gas Insulated Switchgear (GIS)

GIS equipment of different voltages shall be housed in separate rooms with size suitable for the following minimum requirement. However, the same may be modified based on respective GIS manufacturer's recommendation.

Side clearances in GIS Room (on both sides):

	Grid Substation Up To 132kv Voltage	BSP's & Substation With 230kv And Higher Voltage.
Side clearance from main entrance to GIS equipment	4.0	8.0
Side clearances other end to GIS equipment	2.0	5.0
Front Clearance between GIS Room building wall edge to the GIS	3.0	4.0
Rear Clearance between GIS Room building wall edge to the GIS	2.0	3.0

Equipment layouts shall conform to the general guidelines specified in 32-TMSS-02 for the space requirements. For the application in any specific project, space requirement per layout drawings shall govern. Overhead traveling crane of suitable rating shall be provided in the GIS building for maintenance purposes per clause 7.8 of this standard.

5.11 Fire Suppression and Detection Equipment

Fire suppression and detection equipment shall be installed per requirements of TES-B-106, TES-P-119.21 and relevant NFPA standards.

6.0 HEATING, VENTILATION AND AIR CONDITIONING (HVAC) SYSTEMS REQUIREMENTS

6.1 General

6.1.1 Heating, ventilation and air conditioning system shall be provided for the Substation Buildings to maintain all year-round desired indoor temperature and environmental condition for efficient equipment operation. Design and installation shall meet the requirement of TES-K-100.01 and TCS-K-100.01 respectively.

6.1.2 Each substation shall be provided with air conditioning system per TES-K-100.01 and Material standards 73-TMSS-01, 73-TMSS-02 & 73-TMSS-03. Substation building shall be provided with two (2) identical air conditioning units with a common ducting system and having cooling, humidification and heating functions. Each unit shall have 100% cooling capacity and the selected air conditioning stand-by unit shall automatically operate when the other unit is out of service. System selection and evaluation shall follow recommendation set forth by TES-K-100.01.

6.1.3 Chilled or Direct Expansion Split Type Air Conditioning System, a separate mechanical room shall be provided to house both the air handling units and HVAC system control panel. For PACU roof installation, special care to be taken to design the roof accordingly.

6.1.4 Air distribution system shall be through externally insulated rectangular and circular ductwork with all the required air devices and accessories.

6.1.5 The fresh air intake shall be connected to the main return duct and shall consist of the required component such as; sand trap louver, 50mm washable aluminum pre filter, bag filter, manual and motorized volume dampers.

6.1.6 Winter heating and dehumidification shall be through duct-mounted electric heaters to maintain the desired indoor relative humidity.

6.1.7 Temperature and humidity indicators shall be installed at convenient locations in all rooms except toilet, including one outside the building.

6.1.8 Controls shall be direct digital control type. Interlock controls shall be provided with SCADA and FPSP to remotely monitor status of the air conditioning unit and to trip the air conditioning unit in case of smoke/fire detection, respectively. Likewise, provision shall be made not to trip air conditioning unit operation on manual checking of alarm.

The control system shall have provision for management stations and/or service terminal for future connections.

6.1.9 All air handling units, air-cooled condensers, package air conditioning units, air cooled chillers and any other equipment to be installed outside shall be mounted on a reinforced concrete pad, at a minimum of 100mm above the floor or 150mm above the surrounding ground finished grade level with vibration isolators. For PACU, roof mounting to be considered if there is space restriction around the Substation or an optimized air ductwork distribution design is not feasible for ground mounting.

6.1.10 Battery room shall be provided with a centrifugal direct/belt driven type exhaust fan having capacity to evacuate the gases during normal trickle charge, equalized charge and boost charge of the battery. Wall mounted, acid resistant and explosion proof exhaust fan motor, capable of providing at least 12 air change per hour. Exhaust fan must be tripped in the event of fire alarm system activation. The exhaust fan shall bear AMCA certified rating seal for both sound and air performance in accordance with AMCA 211 and AMCA 311. Speed controller shall be provided for proper air balancing. For battery room HVAC air inlet control shall be by means of adjusting volume dampers.

6.1.11 Exhaust fan for toilet shall be of centrifugal direct drive type, wall mounted, complete with front guard and backdraft damper. Exhaust fan control shall be arranged such that the fan starts when the light is switched-on and continue to run for 30 minutes, after the light is switched-off.

6.1.12 Remote On/Off auto toggle switch shall be provided to activate each chiller automatically/manually. The remote switch shall be located on the central plant panel at the mechanical room.

6.1.13 Control of chilled water for Air-Handling Unit (AHU) and Fan Coil Units shall be through electronic three-way valve which responds to electronic thermostat via DDC.

6.1.14 Chilled water piping, concealed or exposed, shall be identified with plastic tape pipe markers. Tags on piping shall be used to indicate service, flow direction and pressure.

6.1.15 Motorized fire/smoke dampers and combination fire and smoke dampers, UL Listed with 3 hour fire rating, shall be provided at locations where ducts and outlets pass through fire rated walls, partitions and floors. Photoelectric smoke detectors shall be provided in the HVAC main supply and return ducts. Perimeter mounting angles, sleeves, breakaway duct connections, corrosion resistant springs, bearings, bushings and hinges shall be installed wherever required. All Motorized fire/smoke damper and combination fire and smoke dampers shall be equipped with “manual override” switch.

The HVAC Systems should be designed to shut down in the event of a fire to prevent the spread of smoke throughout the building.

6.1.16 Balancing dampers shall be provided on duct take-off to diffusers, and grilles and registers, in addition to the dampers specified as a part of the diffuser, or grilles and register assembly.

6.1.17 Air conditioning main control panel shall have 7-day change-over timer adjustable from 1 day to 7 days, to automatically switch on the standby unit and vice-versa. In addition, fault change over shall be provided to start the standby air conditioning unit, if running air conditioning unit fails.

6.1.18 Combined 300 mm dial, wall-mounted thermometer/hygrometer shall be provided in each room in the substation.

6.1.19 Mechanical/rotating equipment including electric duct heater shall be provided with individual disconnect switches.

6.1.20 Positive pressure of approximately 5.0 mm of water shall be maintained within the building.

6.1.21 “Emergency lighting system” near the units control panels shall be provided.

6.1.22 Electromechanical room for HFC- 227 ea (FM200) shall be provided with exhaust fans that shall run continuously to create negative pressure if specified in SOW/PTS. In case of fire, the fans should stop.

6.1.23 Provide low pressure alarms if specified in SOW/PTS on the HVAC DDC panels to avoid ingress of the dust inside the substation for each zone.

6.2 HVAC Design Parameters

HVAC design parameters shall be per latest revision of TES-K-100.01.

6.3 HVAC Control System

6.3.1 The direct digital control system shall also include all the necessary pressure and temperature control protective devices as well as control relays consisting of, but not be limited to, the following in addition to the requirements defined in 73-TMSS-01;

a. Low pressure stat with automatic reset, opens control circuit if suction pressure drops below setting to prevent loss of charge and evaporator coil freeze-up.

b. High pressure stat with automatic reset; shuts off the unit if discharge pressure rises above setting, to protect unit from excessive condensing pressures and to prevent overheating of the compressor.

c. Inherent fan motor protection (temperature sensitive) with automatic reset; opens power circuit to fan motor. Fan motors shall also be protected by factory-installed circuit breakers.

d. Internal thermostat with automatic reset embedded in compressor windings; opens control circuit when sensing excessive temperature rise due to current or motor overloading.

e. Current overload protection (current sensitive and temperature compensated) with automatic reset; opens control circuit if compressor current exceeds limit.

f. Circuit breakers (manually reset) to provide 3-phase current overload protection; shuts off compressor and control circuit when current exceeds trip value.

g. Discharge line thermostat with automatic reset, opens control circuit when discharge gas temperature exceeds limit, to protect against excessive discharge temperatures at low load conditions.

h. Oil pressure switch (manually reset), opens control circuit if oil pressure does not rise to switch setting after compressor starts. Compressor oil level low indication/alarm shall be provided in the control panel.

i. Pressure relief valve (spring loaded), relieves system pressure in case of fire or other cause of overheating.

j. Discharge line check valve to prevent refrigerant migration to evaporator through compressor during off cycle.

k. A positive-acting timer device which shall prevent the compressor from restarting for a minimum of 10-minute period if the power supply is interrupted, to prohibit short cycling.

l. Ammeter and voltmeter for each compressor motor, air handling unit (AHU) motor and electric duct heater.

m. "TRIP" and "RUNNING" indicating lights for each fan and compressor motor.

n. "OPEN" and "CLOSED" position indicating lights for each motorized damper.

o. "ON" and "OFF" indicating lights for heaters.

p. Control switches and indication lamps for all exhaust fans.

q. Fault auto-change of approximately 5 minutes time delay to transfer to standby unit in case of failure.

r. Time delay (adjustment from 80 to 250 seconds) for AHU fan to start when motorized dampers for the selected units are in fully open

positioned. Motorized damper end-switch / limit switch shall be interlock with the AHU fan effectively.

s. Humidistat for controlling room humidity.

6.3.2 Pre-piped gauge board shall be provided with pressure gauges for suction and discharge refrigerant pressures, and oil pressures for each compressor. Valves shall be provided for all gauges. Pressure gauges shall be glacier filled type.

6.3.3 Low voltage, adjustable thermostat to control heating stages in sequence with delay between stages, compressor stages and supply fan to maintain temperature setting shall be provided.

6.3.4 Electric solid state, microcomputer-based room thermostat with remote sensor shall be provided. Thermostat shall be installed with transparent protective cover.

Room thermostat shall incorporate:

a. Automatic switching from heating to cooling.

b. Preferential rate control to minimize overshoot and deviation from set point.

c. Setup for four (4) separate temperatures per day.

d. Instant override of set point for continuous or timed period from one hour to thirty-one (31) days.

e. Short cycle protection.

f. Programming based on weekdays.

g. Switch selection features, including imperial or metric display, 12 to 24-hour clock, keyboard disable, remote sensor, fan on/auto.

6.3.5 Room thermostat display shall include:

a. Time of day

b. Actual room temperature

c. Programmed temperature

d. Programmed time

e. Duration of timed override

f. Day of week

g. System mode indication: heating, cooling, auto, off, fan auto, and fan on.

h. Stage (heating or cooling) operation

6.3.6 Room humidistat shall include the following:

a. Set point range of 20% - 80% RH (Relative Humidity), adjustment shall be external.

b. Built-in temperature measuring element.

c. Switching differential shall be fixed at 4% RH.

d. Removable transparent set point knob cover. When the cover is mounted, the selected set point can be read but cannot be adjusted externally.

7.0 CIVIL/STRUCTURAL REQUIREMENTS

7.1 Building Design Criteria

The parameters and criteria specified below shall be adopted wherever applicable to the basic design of the building and structures, and shall also serve as the basis in the development of the detailed construction design subject to necessary adjustments and implementations with particular consideration of geological conditions of the site base on geotechnical reports and, mobile substation load or wheel load as per AASTHO requirements.

Loading values given below shall be considered as the minimum. Any discrepancies between these values and that of relevant standards and codes shall be resolved in writing prior to the design and engineering.

Design Loadings

7.1.1. Dead Loads

a.	Water	9.87 kN/m³	
b.	Concrete	23.35 kN/m³	
c.	Steel	77.08 kN/m³	
d.	Soil	17.65 kN/m³	(For preliminary purpose)*
e.	Sand	19.61 kN/m³	

Weight of fixed service equipment's including PACU units and Crane dead weight shall be taken into consideration.

* Actual value as recommended by soil report shall be considered during detailed design.

* Other Dead Load to be carried by the Substation Building shall be referred to ASCE/SEI-**7**. Catalogue of materials shall be requested to be submitted for Dead Load not listed in ASCE-7.

7.1.2. Live Loads

The minimum live load values used for structural calculation shall be as follows:

a. Roof

No Access	:	1.0 kN/m²
With Access	:	1.5 kN/m²

b. Floor

(Including Staircase flight and landings)	:	4.8 kN/m²

* Live load at Control & Communication Room shall also be considered as follows:

Concentrated Load	5.0 kN (minimum)
Distributed Load	7.5 kN/m^2 (minimum, including floor, maintenances and forklift live loads)

* Live Load at GIS Room shall be considered as follows:

Distributed Load	12.0 kN/m^2 (minimum, including floor, maintenances and forklift live loads)

* Live Load at Switchgear Room shall be considered as follows:

Distributed Load	10.0 kN/m^2 (minimum, including floor, maintenances and forklift live loads)

Equipment load shall be obtained from the Equipment Manufacturer.

Equipment load shall be designed for the heaviest intensity of equipment load. Impact load factor due to dynamic and operating loads shall be per ASCE/SEI-7.

* For basement floor, live load shall be taken as 4.8kN/m^2

7.1.3. Wind Load

The design wind loads for the building as a whole or for individual components and cladding thereof shall be in accordance with the latest revision of 01-TMSS-01 and/or ASCE/SEI-7, with a basic wind speed of 150 km/h and exposure “C” category. Moreover, Importance factor of 1.15 shall be adopted in calculating wind load.

7.1.4. Seismic Load

The lateral seismic load shall be computed as per 01-TMSS-01 and parameters based on KACST's seismic zone category on various regions of Saudi Arabia. However, in the event that the structure location is located in Zone 0, the applicable seismic zone shall be Zone 1 category. Moreover, Importance factor of 1.25 shall be adopted in calculating seismic load.

7.1.5 Soil, Hydrostatic and Surcharge Pressure

In the design of substation basement wall and vertical structures below grade, provision shall be made for the lateral pressure of adjacent soil. In addition, minimum of 10 kN/m^2 or HS 20 wheel load, whichever is higher, shall be made for surcharge from fixed or moving loads. When a portion or whole of the adjacent soil is below a free water surface, computations shall be based on weight of soil diminished by buoyancy, plus hydrostatic pressure.

7.1.6 Load Factor and Combination of Loads

Substation structure shall be designed for the worst combination of dead, imposed and equipment loads with temperature, wind, seismic, and other effects. The load factor and load combinations to be considered as basis of design shall conform to the applicable codes of ASCE, ACI, and IBC.

7.2 Substation Building

7.2.1 Foundations

Foundation types for Substation building structures and outdoor equipment shall be based on the recommendations of the soil investigation agency as described in reports and as per good engineering practice and requirements.

7.2.2 Cable Basement/Cable Entry room

Cable basement/Cable Entry Room shall be provided at all Substations with regard to water table. Interior concrete surfaces shall be coated with two (2) coats of cementitious elastomeric impermeable membrane after thirty-five (35) days of concrete placement. The second coat shall be applied within the time interval (between successive coating application), specified by the manufacturer. Total wet film thickness shall not be less than 1mm (minimum). Drainage sump shall also be provided.

a. Provide drainage sump pits measuring 900mm length x 900mm width x 750mm height. The sump shall be formed with reinforced concrete at the lowest part of the basement and shall be covered with an effectively grounded galvanized steel grating. A collecting channel of minimum 75mm wide x 75mm deep shall be formed all around the edges of the basement floor at its junction with the basement walls allowing any water penetrating the walls or openings to run to the sump. The number of sump pits shall be at least one for every 200 m^2 of cable cellar area and 2 sump pits per partition in tunnels.

b. All concrete construction joints below grade shall be provided with approved water stop.

c. Sump pit shall be equipped with dedicated and permanently installed submersible sump pump with electric motor, power supply, float switches, automatic controls and a galvanized steel rising main to discharge onto the paved area surrounding the substation.

The discharge pipe shall be positioned to ensure water discharge does not cause flooding, nuisance or erosion of surrounding property. The end of the discharge pipe shall be provided with a wire mesh to prevent the entry of small animals.

Cable basement/Cable Entry room shall be provided with a permanently fixed 50mm outside diameter steel suction pipe. The lower end of the pipe shall be fixed 150mm above the bottom of the sump floor. The upper end of the pipe shall be built through the external substation building wall 1.0 meter above external grade level. A suitable coupling or adapter with removable cap or plug for tanker, or mobile pump used shall be fixed to the upper external end of the pipe. Power supply installation for sump pump shall be carried out as detailed in NFPA 70. A "start/stop/auto" switch with "sump pump power supply on" and "pump running" lamps of all pumps shall be installed on the adjacent wall and l.5m above the entrance to the basement and also on the main control panel. Provide manual controls and monitoring LEDs for all pumps remotely on a central control panel at ground floor of respective building. Only automatic controls will be retained near the pump.

Pump Specification shall be as follows:

Discharge Pipe Diameter	50 mm (nominal)
Discharge Rate	4 Liters per second
Total Head	as required by the installation
Pump Motor	400/230V, 60Hz, Three/single phase(s)

d. Tanking shall be done for all the portions of Cable Basement/Cable Entry Room with waterproofing membrane, extending up to substation site development asphalt finished grade level. All cable/ducts penetrations through basement walls shall be sealed with water stops.

e. On a case to case basis, depending on the environmental surroundings to the substation, underground collecting channel complete with storage or collecting manhole and automatic sump pump with level switch shall be installed outside and around the cable basement at an elevation lower than the basement floor finished elevation.

7.2.3 Ground Floor

The design of slab on grade shall be as per ASCE/SEI-7. Floor of the Slab on grade portion of substation building shall be of concrete with a minimum of 150mm thickness, reinforced with welded wire fabric, deformed steel bars or a combination of both.

For slabs resting on beams (Roof of Cable Entry Room/Basement), minimum thickness shall be 200mm, reinforced with deformed steel rebar. The finished floor level shall have a minimum height of 500-1000 mm above the Asphalt crown elevation or as specified in SOW/TS.

Cable trenches/tunnels formed into the floor slab or false floors shall be installed to provide access to large areas below the finished floor. Floors shall also be suitable to carry the static and dynamic loads of the withdrawable circuit breakers.

Floors shall have smooth steel trowel finish with floor hardener, except at toilet and battery room. Floors shall be coated with dust proof, antiskid and wear resistant floor surfacing. Material shall be epoxy coating 3 or 4 Component applied as per manufacturer's instructions, with minimum thickness of 3.0 mm. Toilet floor shall be provided with anti-slip, ceramic vitrified un-glazed tiles and toilet wall shall be provided with glazed ceramic vitrified tiles from the floor level up to 100mm above false ceiling level. The tiles shall be fixed with adhesives and grouts suitable for damp conditions.

Battery room floor and walls shall be provided with acid resistant, anti-skid, vitrified tiles. Wall tiles shall be extended up to the ceiling. These shall be fixed with chemically resistant epoxy adhesive and grout.

7.2.4 Building Details

Buildings shall be fire resistant, low maintenance, made of material equivalent to Type I-(433) construction as mentioned in TES-P-119.21. All concrete structures shall be designed and constructed in accordance with TES-Q-113.01 and TCS-Q-113.03.

Substation building shall be one or multistoried building, with or without cable basement or cellar underneath the 69/110/115/132/230/380kV switchgears and/or cable termination in GIS or AIS building. Cellar shall have a clear height of 3000 mm from top of basement floor to bottom of ground floor beam. Length and width of cellar shall be sized to provide

sufficient space for pulling, routing and termination of cables to associated switchgears or equipment.

All structural elements shall comply with HCIS/SAF requirement of 2 hours fire resistance.

7.2.5 Roof Slab

The reinforced concrete roof slab shall have a minimum pitch of 20 mm rise for every 1.0 meter of run for proper drainage. The slope shall be provided to the roof structure & shall be in one direction, ie, it shall be from one side and shall not be taken from the center of the roof.

For precast, the pitch of the roof is recommended to be 2% slope for roof with only one direction. However, for cases that require the ridge to be located at the center of the roof, slope shall be 2% minimum on both side with proper consideration on roof drainage.

Roof drains and rainwater drainage piping shall be designed and sized in accordance with Uniform Plumbing Code (UPC). Downspout or Roof drains shall be installed at the designed low point of the roof and equipped with strainers extending not less than 100mm above the surface of the roof slab immediately adjacent to the drain. The rim of the roof drain body shall be at least 5mm below the top of the roof slab.

The slope of reinforced roof slab shall be 2% when PACU is placed on the roof. Fully covered staircase to be provided in cases where PACU is proposed on roof or as mentioned in SOW/TS.

Parapet wall ; For low roof 2.5 meters from top of roof tiles and high roof 1 meter from top of roof tiles.

7.2.6 Septic Tank

a. Hand trim excavation to suit septic tank, distribution box and field tile arrangement shall be provided. Stones, roots or other obstructions shall be removed.

b. Earth work and backfilling around sides of tank shall be done as per TCS-Q-113.02.

c. Installation of septic tank and distribution box and related components shall be provided on bedding.

7.2.7 Metal Doors

All main doors shall be of double-leaf type to permit the entry or removal of the largest piece of equipment in the building. Any door in a means of egress, including door in the battery room shall open outwards and shall swing in the direction of exit travel.

A heavy duty door closer shall be provided to assure that the door will close against the internal positive air pressure. The doors shall include locking devices, astragals and adequate weather & dust stripping pull/push plate, nameplate, surface bolt and necessary hardware such as door stop, door coordinator, and door bumper to permit a rapid exit from the building. All basement doors should be emergency exits, provided with panic device. Recessed double doors can be used after approval by the Company Representative, if the manufacturer does not certify doors with astragals for Fire rating. Main access door or frequently used exits to outside shall be provided with air trap door. All doors shall be fitted with UL listed/labeled fire exit hardware as per NFPA-80 and NFPA-101 and shall be per standard drawing TA-800023, Door Schedule and Details and TA-800009, Steel Doors, Frames, and Hardware for TES-P-119.21 security Requirements.

Motorized shutter shall be installed for main entrance of GIS room if specified in SOW/TS.

Fire rating of hollow metal doors shall be same as walls.

For substation buildings designated as buildings housing "vital or sensitive equipment", typical requirements of doors and hardware for security and emergency exit shall comply with the requirements of TES-P-119.21.

Locks for all doors shall implement a master key system having a minimum of six (6) interchangeable and replaceable pin-tumbler cores except where locks conforming to TES-P-119.21 are required. In other areas, Mastered key pad locked external doors are in use.

7.2.8 Thermal and Moisture Protection

Roof and exterior walls of the building shall be constructed in accordance with standard drawing TA-800024, Finish Schedule/ Miscellaneous Details for thermal protection and waterproofing. The roof and wall shall have “U” factors of not greater than 0.34W/m²-°K and 0.568W/m²-°K, respectively. Roof slab shall be provided with a waterproofing membrane which can be 1 layer of PVC 1.5mm thick or 2 layer of elastomeric bituminous 4 mm thick each, leveling screed, filtration mat, minimum 100 mm thick rigid extruded polystyrene insulation and minimum 50 mm thick washed durable gravel having corn diameter of 15mm to 30mm or concrete tiles of size 400X400X40 mm.

7.2.9 Basement/Cable Entry Room/Trenches/Tunnel Waterproofing

Waterproofing membrane shall be two (2) layers with minimum 4mm thick each of elastomeric bituminous membrane and installed on to the walls and base slab of basement, Cable Entry Room, Trenches, Tunnel, Power cable / control cable trenches and under false floors slabs, walls under control and communication rooms and wherever else required as per SOW/TS. All installations shall be as recommended by manufacturer. The material for

waterproofing shall be forwarded to National Grid Saudi Arabia for review and approval with 10 year guarantee certificate from manufacturer. The waterproofing membrane applied to vertical surfaces shall be protected with a 15mm thick protection board on 2 layers of 4 mm thick Bituminous elastomeric Membrane. Horizontal surfaces shall be protected with a 50 mm thick cement concrete screed and 2 layers of 150 micron polyethylene sheet.

7.2.10 Canopies

Canopies shall be provided over all exterior doors to protect the doors from rain and direct sunlight. The canopy shall be installed at a suitable height above the doors to accommodate lights under it. Top of the canopy shall be sloped to drain water. Minimum projection of the canopy shall be 1200mm.

7.2.11 Access Cage Ladder to Roof

The access to roof shall be at least through the cage ladder. Access ladder with more than nine (9) meters in height shall be provided with resting platform as detailed on standard drawing TA-800049, Access Cage Ladder. Access ladder shall be provided to reach all roof levels.

7.2.12 Suspended False/Drop Ceiling

Suspended False/Drop Ceiling shall be done only if specified in SOW/TS. The details for False/Drop Ceiling, if specified, are as follows;

a. The suspension system shall be an exposed one/two directional metal grid system consisting of painted galvanized iron components with enamel finish.

b. The main runners shall be 40mm high screw slot Tee at 610mm centers. Special hanger unit will slide into the upper rib of the Tee runners and will be spaced at 1200mm centers. Tee runners will be supported with 3.3mm² galvanized wire hangers attached to the structural roof systems.

c. Wall angles shall be fixed to the wall at 300mm spacing.

d. Cross bracing painted galvanized iron Tee will be fixed on top of the main runners and connected to it at 1220mm centers to ensure lateral rigidity of the suspension system.

e. Acoustic tiles shall be mineral fiber, 600mm x 600mm x 20mm size or 600mm x 600mm x 30 mm thick asbestos free type and fire rated for 90 minutes shall be used as may be appropriate. The tiles shall be factory finished in standard vinyl paint and plastic spray. At wet locations finish shall be moisture resistant.

f. Acoustical tiles performance shall be as follows:

i.	Noise Reduction Coefficient (NRC)	:	0.55 - 0.65
ii.	Sound Transmission Class (STC)	:	35 - 39
iii.	Flame Spread Rating	:	25
iv.	Light Reflectance	:	LR-1
v.	Smoke Developed Rating	:	Less than 50

g. Installation of acoustical ceiling system shall be in accordance with manufacturer's recommendations and ASTM C636. For rooms protected with fire protection system and provided with suspended false/drop ceiling, fixing shall be to minimize tiles movement from their grid due to positive gas pressure.

h. The materials for the acoustic ceiling system shall conform to the following standards:

ASTM A167, ASTM B209, ASTM C423, ASTM C635, ASTM C636, ASTM E84, ASTM E90, ASTM E119.

i. The lighting fixtures shall be 600X600 mm or 600X1200 mm.

7.2.13 Passageways

Passageways shall be without any obstruction with minimum headroom of 2400 mm. Passage width for emergency exits and aisles shall be minimum of 1500mm. Indoor busses directly above the passageways shall be protected by metallic safety barrier which shall be grounded properly.

7.2.14 Emergency Exits

At least two emergency exits shall be arranged or located in such a manner to permit escape from the building and basement. One of the exit stairways from the basement shall be used as entrance from switchgear room to basement and cable entry room. In other areas, access to the basement is through an internal access, a stair. A separate door and stair shall be provided outside the switchgear building to allow access to the basement.

7.2.15 Stairways and Steel rails

The stair steps shall be of minimum 1.5 meter width; and rise and run as per IBC. Stairs shall be made of concrete and steps shall be provided with anti-slip surfacing and Pre-formed aluminum nosing, fixed with minimum of 3 flathead expansion bolts. In substation with limited space reduction of the minimum width requirements of 1.2 meter shall be allowed. Stairways shall be provided with handrails complying with the requirements of NFPA101, Life Safety Code. Guardrails above stair opening in the floor slab shall be provided. Handrails and guardrails shall be painted canary yellow.

Steel rails and all types of metallic inserts and embedment for transformers and capacitor banks shall be 130 micron DFT hot-dip galvanized.

7.2.16 Cableways

All cable/duct penetrations including cable tunnel through fire rated walls, floors and ceilings shall be provided with fire stops in accordance with the latest revision of TES-P-119.21. Cable routing can be accomplished by using any of the several methods described below:

a. Cable Trench

Cable trenches for power, control and communication cables shall be formed into concrete floor slab and covered with checkered metal-plate covers, flushed with the finished floor. Preferably the cable trench shall be located adjacent to the control/relay and equipment panels to facilitate panel interconnections. HV and LV power and control cable trenches shall be designed per TES-P-119.20.

b. False Floor (Raised Cavity floor)

i. When large open area is required for cable routing, false floor shall be used. The top of the removable floor panels shall be flush with the finished floor. In some cases, lightweight removable floor panels installed on the adjustable pedestal are positioned in areas requiring extensive cable interconnections or where future plans dictate a large amount of cable rerouting. False flooring shall have a minimum vertical clear space of 900 mm below the floor. The false floor shall be designed for the maximum anticipated rolling and concentrated load of equipment and personnel.

ii. The Communication and Control Room (raised cavity) floor must be leveled and finished with standard tiles.

iii. Raised floor tiles shall be as per 78-TMSS-12, Rev. 0.

iv. The understructure shall be bolted rigid grid 6/2 configuration complete with stringer covers, fasteners, pedestal heads with threaded studs, nut/locking collars, tube/base plates pedestal adhesives and other understructure hardware.

v. One floor panel lifter and one spare shall be provided.

vi. Floor panels shall be interchangeable within the system except if panels are cut for special conditions.

vii. Floor panels shall be mechanically locked trim edge to help prevent loose, shifting or broken trim.

viii Sufficient lighting shall be provided inside false floor for maintenance purposes.

c. Conduits

This method is useful for cable routing in floors or along walls and for cable entrance in the control building. Conduits may be used for wire containment to convenience outlets, lighting fixtures and other control room auxiliary power equipment per TES-P-119.25. Communication conduits for outside plant shall be in accordance with standard drawing TB-800079, PVC Conduit Installation Concrete Encased (Vital Communication).

d. Cable Tray

i. Communication cable trays inside the communication room, when specified, shall be 305mm wide communication type in parallel and at right angles to building walls. Trays shall be located above the equipment. Clearances around or between trays shall be provided to permit adequate access for installing and maintaining the cables. Communication cable trays shall be installed at a height of 2400mm from the floor. The clearance from the top the cable tray to the ceiling shall be at least 400mm.

ii. HV and LV power cable trays and control cable trays shall be installed per TES-P-119.20 and TES-P-119.21. The cable trays in the basement shall be mounted from the ceiling with proper supports and/or hangers.

7.2.17 Toilet/Hammam

a. General

All substation buildings shall be provided with a toilet/hammam. Water connection shall be tapped from the municipal water supply for the toilet fixtures and other fixtures that require water supply. A roof mounted, 2000 liters capacity storage tank shall be provided to ensure a continuous supply of water with adequate pressure at all times. In case municipal water supply pressure is not sufficient to fill the tank, booster pump shall be provided with by-pass line for filling the storage tank. Pump shall be rated at a maximum filling time of 20 to 30 minutes. Pumps and control accessories shall be protected from rain/sunlight by constructing a metallic shelter. A provision for storage tank-truck filling shall be provided complete with transfer pump with bypass line rated at a maximum filling time of 30 minutes.

Waste, drain and sewage disposal systems through septic tank and seepage pit shall be provided if existing municipal sewer line is not available for connection within 200 meter radius from the property.

Septic tank and seepage pit shall be per standard drawings TA-800134, Septic Tank Installation Details For Non-Traffic Area and TA-800135, Disposal System Layout & Seepage Pit Installation Details respectively.

Design and installation of plumbing system shall conform to Uniform Plumbing Code (UPC) and Uniform Mechanical Code (UMC).

b. Plumbing Piping

Installation

- Installation shall be in accordance with manufacturer's instructions.
- Piping shall be installed to conserve building space.
- Elevations of buried piping shall be provided to ensure not less than 1.00 meter and 0.50 meter of cover for traffic and non-traffic areas, respectively.
- At inverted elevations, slopes for drainage of minimum 2% gradients shall be maintained.
- Means to encase exterior cleanouts in concrete, flush with grade shall be provided.
- Water hammer arresters complete with accessible isolation valve on hot and cold water supply piping to lavatories shall be provided.
- Means to disinfect water line including building plumbing system shall be provided.

c. Connecting Piping

i. Connection of outlet between building sanitary piping and septic tank, between septic tank and distribution box, and between distribution box and seepage pits with a minimum slope of 2% shall be provided.

ii. If tests indicate WORK is not meeting specified requirements, removal, replacement and retest of WORK shall be incorporated.

d. Site Sanitary Sewage System

i. Placement of bedding material at trench bottom; leveling of materials in continuous layer not exceeding 150 mm compacted depth; and compaction to 95% of maximum dry density shall be provided.

ii. Optimum moisture content of bedding material shall be maintained to attain required compaction density.

iii. Installation of bedding at sides and over top of pipe to minimum compacted thickness of 300mm and compaction to 95% of maximum dry density shall be provided.

e. Piping Tests

Test shall be performed on all piping and plumbing systems to ensure that they are absolutely leak free. Test pressures shall be in accordance with the required codes and standards. The test pressures given in Table 19-1 below are typical values.

Table19-1: Hydro Test Procedure/Information

Material	Service	Test Pressure	Test Time	Test Fluid
PVC	Gravity Sewer	3meter head or MH Full	24 hrs.	Water
PVC, CPVC	Building Water	1034 kPa	4 hrs.	Water

i. Plumbing piping shall include:

- Water Piping, Buried and Concealed

PVC Pipe: per ASTM D1785, Schedule 80 or UPVC SAS 14/15 or CPVC, ASTM F441 Sch. 80, or Equivalent.

Fittings: Same material as pipe form to suit pipe size and end design, in required valves and fittings and other plumbing fixtures.

Joints: per ASTM D2855, solvent weld with ASTM D2564 solvent cement.

- Water Piping, Above Grade and Exposed

Steel Pipe: per ASTM A53, Schedule 40, galvanized or CPVC, ASTM F441 Sch. 80.
Fittings: Malleable iron, per AN/ASME B16.3
Joints: Threaded

- Sanitary Sewer Piping

PVC Pipe: per ASTM D1785, Schedule 40 or UPVC SAS 14/15 equivalent.
Fittings: Same as pipe materials
Joints: Solvent weld.

- Water Piping, Hot Water

Copper Tubing: per ASTM B88, Type K, hard drawn or CPVC pipes per ASTM F441 Sch. 80.

Fittings: per ANSI/ASME B16.18, cast bronze or ANSI/ASME B16.22 wrought copper and bronze.
For CPVC the same material as pipe form and shall suit pipe sizes and end design.

Joints: per ANSI/ASME B32, solder (95% tin-5% antimony), Grade 95TA.
For CPVC, Joints shall be as per pipe manufacturer standard and procedure

Insulation: Flexible, closed-cellular elastomeric in tubular form per ASTM C 534 with a density of 64kg/m^3 and 25 mm thick.

- Flanges and Unions

  Pipe Size 50 mm and below
  Ferrous Pipe: 1034 kPa malleable iron threaded union
  Copper Tube and Pipe: 034 kPa bronze unions with soldered joints
  Pipe Size Over 50 mm

  Ferrous Pipe: 1034 kPa forged steel slip-on flanges, 1.6 mm thick preformed neoprene gaskets.

  Copper Tube and Pipe: 1034 kPa slip-on bronze flanges, 1.6 mm thick preformed neoprene gaskets.

  Dielectric Connections: Union with galvanized or plated steel threaded end, copper solder end, water impervious isolation barrier.

- Gate Valves 50mm and smaller: shall be threaded; taper; cast bronze, body per ASTM B62 or equivalent, 6% zinc that will not corrode in brackish or saline water at 66°C solid wedge, rating 1035 kPa OWG non-shock.

- Swing Check Valves 50mm and smaller: Threaded soft seated bronze body and cap per ASTM B62; swing type, bronze trim per ASTM B62 or better, all wetted bronze parts to contain less than 6% zinc, rated at Class 1035 kPa, non-shock OWG, suitable for saline water.

- Ball Valves

  Up to and including 50 mm: Bronze one piece body, chrome plated steel ball, teflon seats and stuffing box ring, lever handle and balancing stops, threaded ends.

  Over 50mm: Cast steel body; chrome plated steel ball, Teflon seat and stuffing box seals, lever handle, flanged.

- Plug Valves

  Up to and including 50 mm: Bronze body, bronze tapered plug, non-lubricated, Teflon packing, threaded ends.

  Over 50mm: Cast iron body and plug, non-lubricated, Teflon packing, flanged ends.

ii. Plumbing specialties shall include:

- Floor Drains

  Per ANSI A112.21.1, galvanized-cast iron two-piece body with double drainage flange, weep holes, reversible clamping collar and round, adjustable nickel bronze strainer.

  Floor drains for the battery room shall be acid resisting epoxy coated.

- Cleanouts

  Interior Finished Floor Areas: galvanized cast iron, two-piece body with double drainage flange, weep holes, reversible clamping collar, round with scoriated cover in service areas and round with depressed cover to accept floor finish in finished floor areas.

  Interior Finished Wall Areas: Line type with lacquered cast iron body and round epoxy coated gasketed cover, and round stainless steel access cover secured with machine screw.

  Exterior Unsurfaced Areas: Line type with lacquered cast iron body and round epoxy coated gasketed cover encased in concrete flush with floor grade.

  Battery Room: Acid resistant epoxy coated cast iron.

- Hose Bibs

  Bronze or brass, replaceable hexagonal disc, hose thread spout, chrome plated where exposed, with handwheel.

- Sewer Manholes and Cover

  Precast concrete sewer manhole shall be as per standard drawing TA-800137, Sanitary Sewer Manhole Precast Concrete Plan, Section & Details.

  Cover: for Underground Water Tanks, Sewer Manholes, Distribution Box, Septic Tanks, Seepage Pit and all types of dry pits, use Standard Round cast iron with minimum sized pick hole and frame per standard drawing TA-800138, Drainage Sump, Drainage Channel

And Sewer Manhole Cover Details. Use heavy duty cover and frame in vehicular traffic area.

Steps: 20 mm diameter galvanized steel on 300 mm centers.

- Water Hammer Arresters

  Water hammer arrester shall be pre-charged, suitable for operation at a temperature of 73°C to 149°C and maximum 200 psi working pressure.

iii. Plumbing fixtures shall include:

- Western Water Closet

  Bowl: per ANSI/ASME A112.19.2, with flush tank, floor mounted, siphon jet vitreous china closet bowl, with elongated rim, 40 mm top spud, china bolt caps.
  Seat: solid white plastic, open front extended back, self-sustaining hinge, brass bolts, cover.

- Lavatory

  Basin: per ANSI/ASME A112.19.2, vitreous china wall hung lavatory with 100mm high back, drillings on 100mm centers, and rectangular basin with splash lip, front overflow and soap depression.

  Trim: per ASME A112.18.1, chrome plated combination supply fitting with pop up waste, water economy aerator; chrome plated 17 gauge (1.3 mm) brass type P-trap with clean out plug and arm with escutcheon.

- Wash sink

  Bowl: per ANSI/ASME A112.19.3, single compartment 20 gauge (0.9mm) thick, type 304 stainless steel, self rimming with undercoating, 90 mm crumb cup and stainless steel drain, ledge back drilled for trim.

  Trim: per ANSI/ASME A112.18.1, chrome plated brass supply with swing spout, water economy aerator; chrome plated 17 gauge (1.3 mm) brass type P-trap with clean out plug.

- Eastern Water Closet

  Vitreous china toilet wash down, with flush tank, squat close, cast iron with porcelain enameled finish. Foot treads shall be provided for each unit.

- Ablution Faucet

  Personal ablution faucet with squeeze operated self-closing hand held valve flow control lever, including recessed volume control and aerator nozzle, flexible 1.0 meter long stainless steel base hose, chrome plated wall hook, pressure closing wear-free inlet valve with vandal proof housing integral vacuum breaker.

- Urinal

  Per ANSI/ASME A112.19.2 vitreous china, wall-hung washout urinal with integral flushing rim and trap, 20 mm top spud. Flush valve shall be exposed, chrome plated diaphragm type with oscillating handle.

- Eye wash with hand-held hose spray assembly

  Per ANSI/ISEA Z358.1, combination eye and skin wash, free standing with valves and interconnecting piping and universal emergency sign.

  Eye wash: self-cleaning, non-clogging eye wash with instant action, stay open chrome-plated ball valve easily activated by stainless steel push flag, ABS plastic bowl in safety green eye wash receptor, twin soft PVC covered ABS plastic anti surge heads, stainless steel dust cover, control valve, 40 mm diameter supply and waste pipes and fittings.

  Hand-held hose spray assembly: chrome-plated brass spray head with 20mm outer diameter, at least 1.5 meter long heavy duty rubber hose with hanger and self-closing squeeze valve. Hose spray assembly shall be provided along with the eyewash and should operate at maximum pressure of 276 kPa.

iv. Plumbing Equipment

- Electric Water Heaters

  Type: factory assembled and wired electric, vertical storage.

  Performance: suitable for maximum pressure of 1034 kPa.

  Tank: 57 liter glass-lined welded steel, thermally insulated with minimum 50 mm glass fiber, encased in corrosion resistant steel jacket, baked on enamel finish.

  Controls: automatic immersion water thermostat, externally adjustable temperature range from 16°C to 82°C, flanged or screw in nichrome elements, high temperature limit thermostat.

Accessories: brass water connections and dip tube drain valve, high density magnesium anode and ASME rated temperature and pressure relief valve.

- Water Storage Tank

Water storage tank shall be of Polyethylene, layered and insulated per ASTM D1998, horizontal type, manufactured (locally) from polyethylene granules. The tank shall be weather-resistant, chemical-resistant, impact-resistant, and non-toxic and U.V. (radiation) stabilized. The tank shall have uniform thickness of at least 13mm and thermal conductivity of 0.112W/MºK. The inside surface of the tank shall be clean, smooth, uninterrupted, free of foreign material. It shall be properly cleaned and maintained regularly to resist the growth of bacteria, fungi and algae. Capacity of the tank shall be 2000 liters. Ladder shall be provided for access to the manhole cover for maintenance.

Tank shall be equipped with nozzles for water inlet and outlet, drain, float switch and a level sight glass mounted for clearer visibility. Necessary pipe-fittings shall also be included in the supply.

Tank shall also be provided with 600mm x 600mm manhole for maintenance purposes. Manhole cover shall be designed to prevent ingress of sand, dust, rainwater and securely fixed by screws/bolts.

v. Sanitary sewer shall include:

- Septic Tank and Distribution Box

Septic Tank: reinforced concrete construction, similar to Standard Drawing TA-800134, Septic Tank Installation Details For Non-Traffic Area. Piping material shall be PVC.

Distribution Box: reinforced concrete, single inlet, two, gate, removable cover with lift ring construction.

- Connecting Pipe Materials

Plastic Pipe (PVC): per ANSI/ASTM D2729, bell and spigot solvent sealed joints.
Fittings: same material as pipe, tee bends, elbows, cleanouts, reducers, ends to suit pipe joint.

- Seepage Pits

Construction shall be similar to standard drawing TA-800135, Disposal System Layout & Seepage Pit Installation Details.

7.2.18 Workshop/Office

The workshop/office shall be as per the dimensions given in project drawings and SOW/TS with following minimum requirements;

a. Separate rooms shall be provided for maintenance workshop/store and/or office/record space, if specified. For office, space requirements shall be limited to a desk and at least two (2) chairs, one filing cabinet to hold one set of operation and maintenance manual, and drawing chest to hold one set of as-built substation drawings. Office furniture shall be of non-combustible materials.

b. If maintenance workshop is required, it shall be constructed adjacent to metalclad switchgear room and it shall include sufficient work space, a work bench and a tool rack. The maintenance workshop shall be equipped with AC outlets per TES-P-119.30. A common access door shall be constructed between metalclad switchgear room and workshop. A hook at the ceiling of the workshop shall be installed so that chain pulley block can be used for maintenance purposes.

7.2.19 Painting

Painting and color of exterior of the building shall be in accordance with National Grid Saudi Arabia standard TES-H-107.01. Precast concrete structures shall be with exposed aggregate finish.

7.2.20 Building Assemblies/Members Fire Resistance

The structural members shall be fire resistance and minimum concrete cover shall be as per the requirements of ACI 216.1M.

7.3 Roads and Walkways

7.3.1 Paved access roads in general shall be provided in substations where adequate access is required for heavy equipment or vehicles such as cranes, large trucks, trailers and oil-filtering equipment or mobile transformers, under all weather conditions. Maximum grade on the access road shall not exceed 7% so that heavy transformers may be transported by normal movers without any problem. For short distances of 60-100 meter, 10% grade may be permitted. Curvature shall also be taken into consideration. Access roads inside radius at 90 degree intersection shall not be less than 15 meter in order to provide sufficient turning space for long vehicle. Access road width shall be 6 meters, crowned at the center for drainage. Asphalt Concrete paving over an aggregate base course shall be done. The thickness of the asphalt paving and the aggregate base course shall be designed to withstand the maximum anticipated wheel and/or axle load, based on the substation site soil properties. Minimum thickness of asphalt paving or access road shall comply with the requirements of TCS-Q-113.01.

7.3.2 Access road edges or kerbs shall be painted with reflectorized Zinc yellow and Traffic Black paint in stripes. The center line of road shall be painted with reflectorized white broken line. Road traffic warning signs shall be provided at suitable locations.

7.3.3 Walkways shall be designed for a live load (same as substation building floor) of 200kg/m² minimum or a single concentrated load of 150 kg occupying a space of 305 mm square. Concrete sidewalk shall be provided around the perimeter of the building. Width of the sidewalk shall be 1500mm minimum from the edge of the building exterior wall which shall be provided with control joint of 6mm x 25mm depth at every 4000 mm center to center distance and expansion joint at every change in directions and shall be at the same elevation of the floor level of the building. Loading platform shall be designed for a live load of 4.8 kN/m^2 and shall be of minimum width of 2500mm shall be provided at all equipment doors.

7.4 Site Development

The finished grade (crown) of the substation site development shall be at least 1.0m above the highest asphalt level or grade level of the nearest adjacent paved or municipal road; or minimum 2.0m above the highest existing plot ground level; or bottom of the base slab of basement floor shall be at least 1.0m above the ground water level, whichever gives higher elevation.

7.4.1 Substation site development shall be graded to drain water away from the building towards the perimeter fence/boundary wall. The substation site development includes surface paved with asphalt in vehicular traffic areas and 50 mm sized gravel in non-traffic areas of the Substation to prevent wind erosion or deposits of drifting sand. The particulars of asphalting areas and non-asphalting area shall be as specified in Scope Of Work/Technical Specifications (SOW/TS) and conceptual drawings of the project.

7.4.2 The manhole/handhole shall be designed to prevent the entry of water or sand into it.

7.4.3 In the yard areas which are not accessible to compactor/asphalt paver, or narrow and long portions, provide 200 mm thick concrete pavement with double caged steel mesh.

7.5 Fencing and Boundary Wall

7.5.1 Fencing shall comply with the requirements of National Grid Saudi Arabia Standard 70-TMSS-02.

7.5.2 Precast boundary wall shall be provided for substations in urban areas. Precast wall panels shall be constructed as per Standard Drawing TA-800012, Precast Concrete Boundary Wall (Plan, Elevation & Details) and TA-800144, Precast Boundary Wall Type-II (Typical Layout Plan, Elevation, Sections And Details). The design of the wall shall be as per the guidelines of

Ministry of Interior (High Commission for Industrial Security) and reviewed by concerned departments as per standard TES-P-119.21.

7.5.3 For indoor substations, boundary wall facing the power transformers shall be removable precast panels to provide easy access for installation and maintenance. The precast panels shall be constructed as per Standard Drawing TA-800012, Precast Concrete Boundary Wall (Plan, Elevation & Details) and TA-800144, Precast Boundary Wall Type-II (Typical Layout Plan, Elevation, Sections And Details). For urban area substations and vital substations removable precast panels shall also be provided.

7.5.4 All metallic portions of the boundary wall, namely extension arms, drive gate and personnel gate shall be constructed per TES-P-119.21. The perimeter fence/boundary wall shall be provided with warning signs per TES-P-119.21.

Minimum clearance between the removable fence and transformers shall be 4.2 meters. Minimum distance between the boundary wall or removable precast panels/fence and the edge of the fire barrier shall be 3.5 meters. If, the clearances are not possible due to size of the site, these shall be as specified in SOW/PTS.

7.5.5 All metallic portion of fence and boundary wall including gates shall be grounded per TES-P-119.10.

7.5.6 The location of fence/boundary wall and gates shall be per National Grid SA approved drawings.

7.5.7 National Grid Saudi Arabia monogram per standard drawing TC-800003, Signboard on Substation Building shall be installed at the exterior face of the gates and boundary wall/fence.

7.6 Substation Signboard

7.6.1 Before construction commences, Project Signboard shall be provided and constructed on site as per Standard Drawing TC-800003, Signboard on Substation Building.

7.6.2 The substation shall be provided with a signboard per standard drawing number TC-800003, Signboard on Substation Building. The signboard shall be fixed on the external side of the building, facing the main drive gate. The substation shall be provided with a signboard at Gate House as per standard drawing TA-800111.

7.6.3 Before fabrication commences, the Contractor shall submit the following for review and acceptance by the Company:

a. Large scale drawing of the signboard faces showing both the English and Arabic title, and logo.
b. Detailed fabrication drawings.
c. Description of materials to be used.

7.7 Protective Crash Barriers

Protective crash barriers, if required, and specified in SOW/TS, shall be removable and provided with metal chains per standard drawing number TA-800133, Removable Crash Barrier, Location Plan and Detail in Substation. Crash barriers shall be made of 150mm diameter Sch. 80 steel pipe (minimum 1.0 meter height) conforming to ASTM A53, Grade B. The center-to-center spacing between the crash barriers shall be 1.6 meters. The crash barriers shall be installed at a minimum distance of 0.5 meter from the equipment. In case of power transformer the crash barriers can be installed along the edge of the fire barrier wall, away from the equipment. Pipe and metal chain shall be painted with 100mm wide canary yellow and black stripes (alternate pattern). Pipes and chain shall be painted with two coats of zinc chromate and two coats of high gloss durable weather resistant enamel paint. Pipe shall be connected to substation grounding.

7.8 Electric Overhead Traveling Crane in GIS Buildings

7.8.1 General

a. Supply and install an electrically operated traveling crane system including accessories required to complete the system in the GIS and AIS buildings. The unit shall be single/double-girder design with top running bridge, top running electric operated trolley hoist and equipped with floor control pendant mobile along the crane bridge.

b. The minimum capacity of the crane shall be 5 tons, or based on manufacturer's recommendation for the heaviest single component to be installed in the building. The capacity shall be conspicuously marked on the body of the crane as per Occupational Safety and Health Administration (OSHA).

c. Crane shall be provided with maintenance platform with guard rails for maintenance of lighting fixtures and fire detectors mounted on the ceiling. Clearance from top of platform to bottom of roof beam shall be minimum of 1300mm from the lowest point. Access to the maintenance platform shall be through a permanent installed wall-mounted galvanized steel caged ladder. “Emergency Trip Switch” control in the hand held control box shall also be provided.

   Clearance between bottom of hook of crane and topmost point of GIS equipment, including test busing in assembled condition shall be as per manufacturer recommendation.

7.8.2 Crane Equipment/Accessories

a. Crane girder shall have lifting capacity to handle the heaviest module in the GIS buildings. Girders are to be selected based on minimum of 1/800 of the span with built-in safety factor of 5:1. Electric wire rope hoist shall

be considered subject to lifting small and medium capacity loads in periods of about equal time.

b. Trolley shall be motor driven with creep speed. Wheel bearings shall be sized for a minimum bearing life of 5,000 hours at selected capacity and speed. Gear bearing life shall be minimum 25,000 hours.

c. Installation of hoist shall be for standard headroom. Crane/hoist shall have creep speeds for lifting loads and travel.

d. Crane/hoist shall be floor controlled through multi-button pendant suspended from a carriage which is independently mobile along Crane Bridge. Push button in the pendant station shall return to off when pressure is released.
In case of GIS overhead bay arrangement, remote operated (Wireless) controller shall be provided.

e. Installation shall include protective earthing conductor, thermal overload protection and a main contractor.

The following features shall also be included for safety purposes;

1. Hoist Brake Manual Release during power failure.
2. Yellow flashing light (Minimum 6 inch Diameter) during crane long travel.
3. Alarm/horn with red light during overload.

f. Steel structural parts shall be hot dip galvanized and painted with one coat of primer and one finish coat of enamel for corrosion protection.

g. Span of crane shall be able to mount on the steel girder on the bay provided for the equipment.

h. Power supply for the crane shall be 400 volts, 3-phase, 60 Hz, 4 Wire. Motors shall be per NEMA rated with "Class F" insulation. Thermal detectors shall be provided for the motors.

i. Crane stops or mechanical/electrical limiting device shall be installed on the equipment, rails, tracks, or trolleys, to prevent unit overrunning beyond safe limits.

j. Rated load capacity of the crane/hoist should be plainly marked on each side of the crane and should be clearly visible from the ground.

k. All trolley and bridge drive motors shall be furnished with at least two braking systems.

l. Bridge power conductor system shall be UL approved, multi-conductor flat cable for festoon system suspended and traversing bridge.

m. Bridge control shall be designed for single and two speed bridge operation. Controls shall have NEMA 12 enclosure. Control circuit shall be rated for 230Vac and protected by fuse.

7.9 Jib Crane

Jib crane of minimum capacity of 2 Tons, with Direct Electric Drive and capable of lifting the heaviest single electric component to be installed in the Control Building, Switchgear Building and GIS Building. The maximum span shall be enough to lift the weight from the platform and transfer it to the ground floor easily and vice versa.

The reactions coming from the Jib crane shall be supported by the substation building framing system. The design of the jib crane support and framing system of the substation building shall consider the loads/reactions from Jib Crane and to withstand the load rating to accommodate the heaviest electrical panels or parts of the Switchgear, Control Room and maintenance parts of other electrical equipment.

The Electric Power supply: 400 volt, 60Hz, 3 -phase, 4-wire. In addition, the SLEW movement of the arm shall be electrically operated.

7.9.1 The unit system shall be provided with mechanical limiting device to prevent unit overrunning beyond safe limits and shall be as specified below:

a.	Wall-mounted or Pad Mounted type as required with Electric chain hoist	
b.	Degree of rotation freedom*	Wall Mounted 180 deg. (min.) Pad Mounted 270 deg. (min.)
c.	Minimum working radius	the platform plus largest component dimension
d.	Working Height	as per Manufacturer recommendation
e.	Lift Height / Depth	to be considered for ultra deep truck level
f.	Working Temperature Range	As per 01-TMSS-01
g.	Protection	Hoist - IP55, Traveling - IP55
h.	Electric Power Supply	400-230V, 3phase, 60hz
i.	Motor Insulation	Class F

* Provide PVC stopper at canopy level if there is chance of Jib Crane hitting it.

7.10 Gate House

Gate House, if required in a substation, and specified in SOW/TS, shall be as per National Grid Saudi Arabia Standard Drawings TA-800039, TA-800040, and TA-800041.

7.11 Cable Tunnel

Cable tunnels shall be formed with reinforced concrete top slab, side walls and floor slab. The design of the members shall be as per ACI and relevant Company and International Standards.

7.11.1 The minimum clear width of tunnel shall be five (5) meters and the minimum clear height of the tunnel shall be three (3) meters for 380 kV Substations. For **132** kV Substations, Cable tunnels shall be sized adequately, taking into consideration the following aspects: space for maintenance and operation, all future piping to be installed and all cables installation plus a 20% spare capacity.

7.11.2 Cable tunnels shall be provided with metallic cable trays, galvanized steel supports for the cables and necessary accessories as needed.

7.11.3 A minimum of two (2) emergency exits from the cable tunnels shall be provided. Fire rated doors and reinforced concrete stairways with pipe railing shall be provided at the emergency exits to allow exit from the cable tunnels or entrance to cable tunnel from the substation yard.

7.11.4 The floor slab of the tunnel shall be slope towards the drainage sump pit at 0.5% slope with the use of cement screed.

7.11.5 Sump pit shall be equipped with dedicated and permanently installed submersible pump with electric motor, power supply float switches, automatic controls and a galvanized steel rising main to discharge unto the paved area surrounding the substation.

7.11.6 A minimum of two (2) sump pits shall be provided. The size of the sump pits shall be 900mm x 900mm x 750mm. All sump pits shall be located at the center of the Cable Tunnels or along the walkway route.

7.11.7 Cable tunnels shall be properly ventilated and provided with lighting arrangement.

8.0 LIGHTING AND RECEPTACLE REQUIREMENTS

Outdoor lighting requirements shall be as per TES-P-119.21. Indoor lighting and receptacle requirements shall be as per TES-P-119.25.

9.0 LIGHTNING PROTECTION REQUIREMENTS

Lightning protection of the substation building shall be per TES-P-119.07. Lightning protection for communication facilities shall be per TES-T-111.03.

10.0 GROUNDING REQUIREMENTS

The permanent protective grounding system for substation electrical equipment and communication facilities shall be designed per TES-P-119.10 and TES-P-111.02, respectively. It shall also comply with the applicable requirements of ANSI C2.

11.0 FIRE AND LOSS PREVENTION AND SECURITY REQUIREMENTS

For fire and loss prevention and security requirements, refer to TES-P-119.21, and relevant NFPA Standards. Fire Protection and Prevention Requirements for National Grid Saudi Arabia Facilities and the NFPA 72. Intrusion alarm connected to SCADA via interface cabinet (IFC) shall be provided in the substation building per Scope of Work/Technical Specifications (SOW/TS) of the project. Fire protection of communication facilities shall be per TES-T-111.17. Fire protection and intrusion alarm devices shall be so located such that these can be easily accessible and safely maintained without requiring outage of power equipment.